

architecture featured approximately 65 million trainable parameters.

This explosive growth has reinforced the belief that 'bigger is better,' positioning trillion-parameter models as the benchmark for AI success. However, these massive models are typically optimised for broad, consumer-oriented applications rather than specialised needs.

For enterprises that demand domain-specific accuracy and efficiency, blindly pursuing larger parameter counts can be both costly and counterproductive. The key question is whether a model's scale matches the problem it aims to solve.

Analysing large language models through a technical lens, not marketing spin

A generative AI model's parameter count dictates its hardware compute demands. Even a small LLM, such as a 7-billion-parameter model, can consume approximately 67GB RAM at a low precision of 16-bit representation. Scaling to larger models drives memory requirements into the hundreds of gigabytes, forcing enterprises to deploy high-end GPUs, multi-socket CPUs, or specialised NPUs to sustain inference throughput.

These hardware requirements are not just a procurement issue; they influence the entire operational strategy. Larger model parameters mean more VRAM for GPUs, faster interconnects, and larger on-chip cache demands for NPUs. They also mean higher energy consumption and cooling costs. In short, chasing a trillion-parameter model without a clear use case risks over-investing in infrastructure that sits idle between inference requests.

For many enterprises, the smarter move is to choose right-sized models that balance accuracy with operational efficiency. That is where methods such as retrieval-augmented generation (RAG) and fine-tuning come into play, allowing enterprises to achieve targeted performance without scaling hardware budgets to the breaking point.

Enterprise RAG vs fine-tuning: Choosing the right parameter count

One of the most critical strategic decisions is whether to:

- Fine-tune the model so knowledge is embedded in its weights,
- Or, pair a smaller, generalist model with RAG for dynamic information.

Although ChatGPT-3 was trained on data spanning millennia, banks do not need historical breadth to answer loan or KYC queries. A focused 7-billion-parameter LLaMA, fine-tuned on the bank's manuals, workflows, and local regulations, embeds domain-specific knowledge, delivering higher accuracy on industry-critical tasks than a far larger general-purpose model.

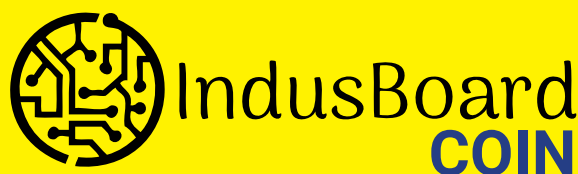
From the perspective of hardware compute requirements, this approach is dramatically more efficient: a 7-billion-parameter LLM model at 16-bit precision



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